

Group 6 Final Report

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ACM Reference Format:

David Espinoza, Christian Cyr, Musama Kuya, and Adam Warniment. 2026. Group 6 Final Report. In *Proceedings of Open Machine Learning Project (CSCI 4420/5420 Project)*. ACM, New York, NY, USA, 6 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Dementia is one of the leading causes of memory loss among individuals aged 60 and older. The most common cause of dementia is Alzheimer's Disease (AD), a progressive neuro degenerative disorder that progressively affects memory, reasoning abilities and cognitive functions. Although there is currently no cure for AD, early diagnosis and proper management can significantly improve the quality of life for patients and families and help slow disease progression[3].

Early detection is important because lifestyle modifications, such as diet, regular physical activity, and medical management of risk factors, can help slow disease progression and reduce symptom severity. However, one of the major challenges in the healthcare system regarding early AD detection is accurately distinguishing between individuals who have already developed AD and those who are Healthy Controls (HC) or have Mild Cognitive Impairment (MCI). Misclassification or delayed diagnosis can lead to inadequate treatment, increased healthcare costs, and a higher psychological burden for both patients and caregivers [2]. Therefore, developing reliable methods for early and accurate classification of disease stages is essential.

Machine learning techniques have emerged as powerful tools for analyzing complex medical data and supporting clinical decision making. By learning patterns from patient features, machine learning models can assist in identifying relationships that are difficult to detect using traditional methods. In particular, classification algorithms have been widely used in healthcare applications to distinguish between disease states and predict clinical outcomes. Previous studies have shown that machine learning models can

achieve strong performance in medical classification tasks and assist in clinical decision making [1].

In this project, machine learning classification models are developed and evaluated to address the problem of Alzheimer's disease detection. The objective is to build a multi-class classification framework capable of distinguishing between cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease (AD) patients based on clinical data. Additionally, the performance of different models is analyzed to understand the impact of model selection and hyperparameter tuning on predictive accuracy and generalization performance.

2 Problem Definition

The problem is formulated as a supervised multi-class classification task for Alzheimer's disease diagnosis. Given an input feature vector $x \in \mathbb{R}^n$, where each component represents clinical or diagnostic information from a patient, the objective is to predict a class label $y \in \{\text{CN}, \text{MCI}, \text{AD}\}$, corresponding to cognitively normal, mild cognitive impairment, and Alzheimer's disease, respectively.

The goal is to learn a function $f(x)$ that maps the input features to the correct class label using labeled training data. The learned model should generalize well to unseen data, minimizing classification error while maintaining robustness to variations in the dataset.

Model performance is evaluated based on classification accuracy and generalization capability, with particular attention to the ability to correctly distinguish between intermediate stages such as mild cognitive impairment and Alzheimer's disease, which represent the most challenging cases in practice.

3 Dataset Description

3.1 ADNI Dataset Overview

This dataset is derived from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a large-scale research effort focused on understanding the progression of Alzheimer's disease using clinical, imaging, and biomarker data. Specifically, the dataset corresponds to a processed subset of the ADNIMERGE table, which integrates multiple sources of patient information into a structured tabular format.

The selected features include demographic variables, lifestyle factors, medical history, and cognitive health indicators such as age, gender, BMI, smoking status, physical activity level, education

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CSCI 4420/5420 Project, Las Cruces, NM, USA

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ACM ISBN 978-1-4503-XXX-X/2026/03

<https://doi.org/XXXXXXX.XXXXXXX>

level, diabetes, hypertension, family history of Alzheimer’s disease, sleep quality, and cognitive assessment scores.

The dataset includes samples from three classes: cognitively normal individuals (CN), subjects with mild cognitive impairment (MCI), and patients diagnosed with Dementia. Each sample includes features such as demographic information (age, gender, education), cognitive test scores (e.g., MMSE, ADAS13), neuroimaging-derived measurements (e.g., hippocampal volume), and selected clinical variables.

The ADNIMERGE dataset integrates these heterogeneous data sources into a single table with a large number of features, allowing for comprehensive analysis of disease progression and classification tasks. This makes it particularly suitable for machine learning applications, where the goal is to identify patterns across multiple modalities and distinguish between different stages of cognitive decline.

3.1.1 Exploratory Analysis. Since multiple observations per patient are available, only baseline measurements were selected to avoid data leakage and ensure a single observation per subject. The dataset also contains a significant amount of missing values, particularly in biomarker-related features such as PIB, FBB, and AV45, which exceed 80% missing data. These features were excluded from the analysis, and a subset of relevant features with lower missing rates was selected for model training. After filtering the baseline data, a total of 2409 valid samples were obtained.

The class distribution shown in Figure 1 indicates that the dataset is moderately balanced, with MCI being the largest class, followed by CN and Dementia. This distribution is suitable for multiclass classification without requiring significant resampling techniques.

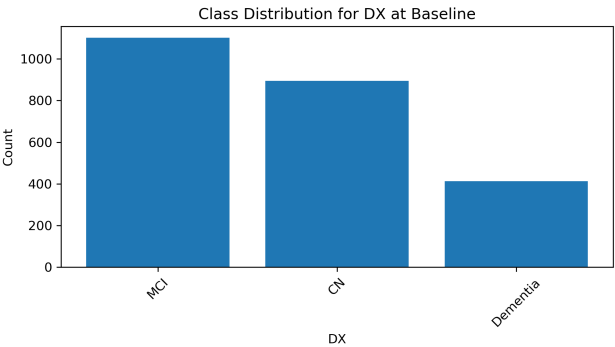


Figure 1: Class distribution of the dataset.

The MMSE and ADAS13 scores shown in Figure 2 and Figure 3 respectively, provide an example of the type of features selected for this study. Both variables show a clear separation between diagnostic classes, with cognitively normal subjects (CN) exhibiting high MMSE and low ADAS13 values, while Dementia patients present the opposite trend. MCI subjects fall between these two groups. This inverse relationship highlights the complementary nature of these cognitive assessments and supports their inclusion as relevant features for classification. This behavior is further supported by the correlation analysis shown in Figure 4, which reveals a strong negative correlation between MMSE and ADAS13, reflecting their

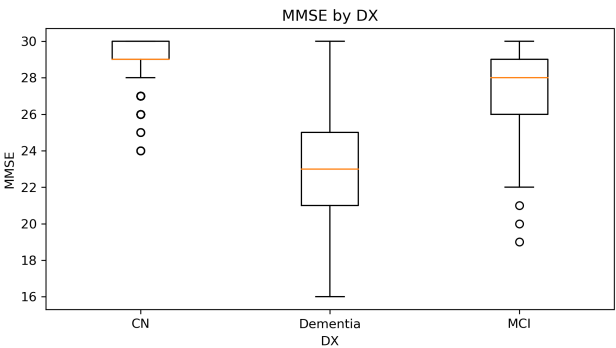


Figure 2: MMSE scores across diagnostic classes.

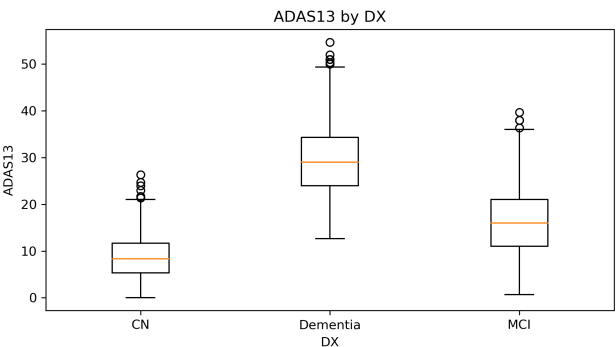


Figure 3: ADAS13 scores across diagnostic classes.

inverse scoring systems. Additionally, volumetric features such as Hippocampus and WholeBrain exhibit positive correlations, indicating consistency in brain structure measurements. These relationships suggest potential redundancy among features and highlight the importance of careful feature selection.

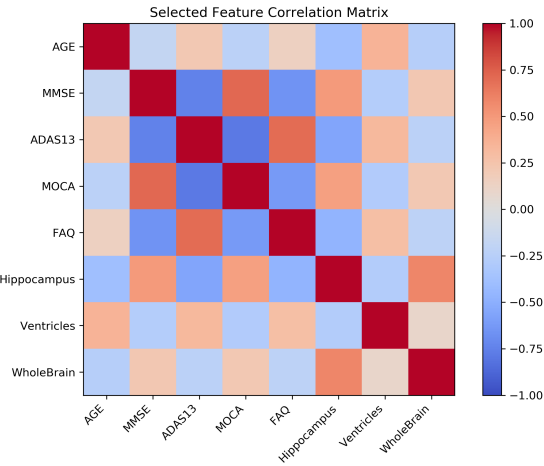


Figure 4: Correlation matrix of selected features.

3.2 Additional Large-Scale Alzheimer's Dataset Overview

In addition to the ADNI dataset, this work examines a large-scale Alzheimer's prediction dataset and a neuro-imaging Oasis dataset. The Alzheimer's prediction dataset contains 74,283 patient records and 25 total features. The dataset includes 24 predictor variables and one target variable, *Alzheimer's Diagnosis*.

The selected features include demographic variables, lifestyle factors, medical history, and cognitive health indicators such as age, gender, BMI, smoking status, physical activity level, education level, diabetes, hypertension, family history of Alzheimer's disease, sleep quality, and cognitive assessment scores.

To ensure consistency across experiments, the same preprocessing and machine learning pipeline developed for the ADNI-based experiments was reused, with dataset-specific modifications applied to feature selection, target mapping, and model evaluation.

3.3 OASIS Dataset Overview

The Open Access Series of Imaging Studies (OASIS-) is an MRI longitudinal data collection from 150 people aged 60 to 96 years, obtained from the Kaggle platform. Within this data set, the same subjects are visited, one year apart. This ensures that the brain atrophy rate is studied in non-demented, demented, and "converted" individuals—those who transitioned to dementia during the study.

Selected features include the Clinical Dementia Rating (CDR) for disease staging, Mini-Mental State Examination (MMSE) scores to measure cognitive impairment, and Normalized Whole Brain Volume (nWBV) a volumetric data to track physical atrophy and Estimated Total Intracranial Volume (eTIV) to normalize anatomical brain size differences.

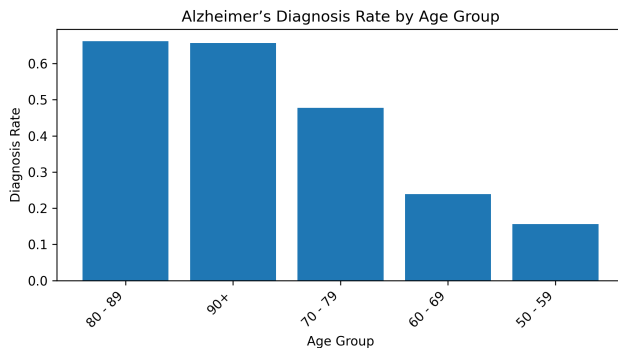


Figure 5: Alzheimer diagnosis rate across age groups in the large-scale dataset.

Oasis Dataset

3.4 Preprocessing

The preprocessing stage was designed to generate multiple candidate datasets for subsequent model comparison, while preserving clinically meaningful information. Since the baseline dataset still contained missing values in several features, a structured preprocessing pipeline was applied.

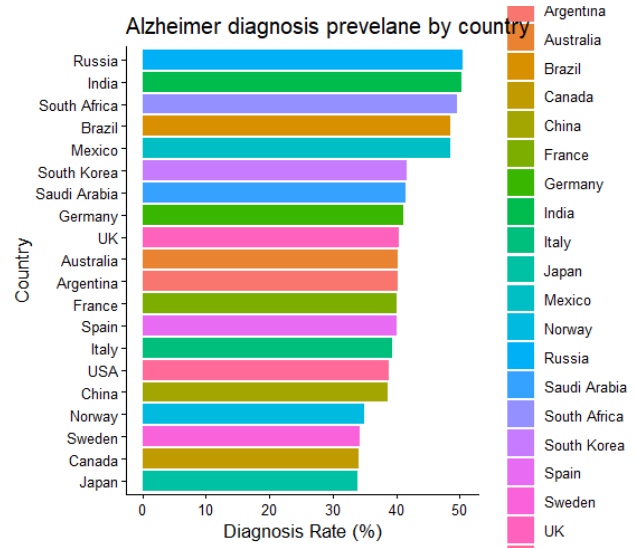


Figure 6: Diagnosis prevalence across countries in the large-scale dataset.

First, non-predictive columns were excluded from the analysis. These included patient identifiers, visit and date fields, acquisition metadata, and auxiliary baseline variables that do not contribute meaningful diagnostic information. This step ensured that the model inputs were restricted to relevant demographic, cognitive, and volumetric features.

Two feature selection schemes were then considered. The first scheme used a manually selected subset of clinically relevant variables. This approach preserves important biomarkers, but introduces a higher proportion of missing values. The second scheme followed a data-driven approach, retaining only features with less than 10% missing values after column filtering.

To handle missing data, two strategies were applied for each feature selection scheme. In the first approach, missing numerical values were imputed using the mean of each feature, while missing categorical values were replaced with the most frequent category. In the second approach, rows containing missing values were removed entirely. These datasets were retained to provide multiple options and determine which configuration performs best with the selected models.

4 Methodology

A diverse set of machine learning models was selected to evaluate their performance on the classification task. The selected models include linear classifiers, distance-based methods, tree-based models, and ensemble techniques. This selection allows comparison between simple and complex approaches, as well as between individual learners and ensemble methods, in order to analyze their ability to capture patterns in the dataset.

Several of the implemented models, including Adaline, Stochastic Gradient Descent (SGD), Logistic Regression, and Linear Support Vector Machines, are based on a linear decision function of the

form:

$$z = w^T x + b \quad (1)$$

Where w is the weight vector and b is the bias term. Classification is performed by applying a threshold at zero:

$$\hat{y} = \begin{cases} 1 & \text{if } z \geq 0 \\ -1 & \text{otherwise} \end{cases} \quad (2)$$

4.1 Adaline (Batch Gradient Descent)

Adaline minimizes the Mean Squared Error (MSE) cost function:

$$J(w) = \frac{1}{n} \sum_{i=1}^n (y_i - z_i)^2 \quad (3)$$

Where n is the number of training samples. The parameter updates are:

$$w := w + \eta \frac{2}{n} X^T (y - z) \quad (4)$$

$$b := b + 2\eta \cdot \text{mean}(y - z) \quad (5)$$

The training curve corresponds to the Mean Squared Error recorded at each epoch.

4.2 Stochastic Gradient Descent (SGD)

The SGD implementation updates the weights using one training sample at time. For each sample, the update rule is:

$$w := w + 2\eta (y_i - z_i) x_i \quad (6)$$

$$b := b + 2\eta (y_i - z_i) \quad (7)$$

The loss for each sample is defined as:

$$(y_i - z_i)^2 \quad (8)$$

And the mean loss over all samples is stored for each epoch.

4.3 Logistic Regression

Logistic Regression is a probabilistic linear classifier that estimates the probability that a sample belongs to a particular class. The linear model output is transformed using the logistic sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (9)$$

which maps real valued inputs into the interval $[0, 1]$. The predicted probability of class membership is given by:

$$p(y = 1 | x) = \sigma(z) \quad (10)$$

A class prediction is obtained by applying a threshold to this probability. The model parameters are learned by minimizing the logistic loss function.

4.4 Linear Support Vector Machine (SVM)

A Linear Support Vector Machine constructs a decision boundary that maximizes the margin between two classes. The classifier searches for the hyperplane that separates the data while maximizing the distance to the closest training samples, known as support vectors. The decision function is defined as:

$$f(x) = w^T x + b \quad (11)$$

The optimal hyperplane is obtained by minimizing the objective function:

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad (12)$$

subject to the constraints:

$$y_i (w^T x_i + b) \geq 1 - \xi_i \quad (13)$$

where C is a regularization parameter that controls the trade off between maximizing the margin and allowing classification errors, and ξ_i are slack variables that permit misclassification.

4.5 Non-linear SVM using Radial Basis Function (RBF) kernel

When the data are not linearly separable in the original feature space, Support Vector Machines can apply the kernel trick to implicitly map the data into a higher dimensional space where a linear separation becomes possible. The decision function becomes:

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \quad (14)$$

where $K(x_i, x_j)$ is the kernel function. In this study, the Radial Basis Function kernel is used:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (15)$$

The parameter γ controls the influence of individual training samples. Larger values produce more complex decision boundaries, while smaller values lead to smoother boundaries.

4.6 Decision Tree

Decision trees select the optimal feature split by minimizing the impurity of the resulting nodes. A common impurity measure is the Gini index:

$$\text{Gini}(t) = 1 - \sum_{i=1}^c p(i | t)^2 \quad (16)$$

where $p(i | t)$ is the probability that a sample in node t belongs to class i . The best split is selected by maximizing the information gain:

$$IG(D_p, f) = I(D_p) - \sum_{j=1}^k \frac{N_j}{N_p} I(D_j) \quad (17)$$

where $I(\cdot)$ represents the impurity measure, D_p is the dataset at the parent node, and D_j are the subsets obtained after the split.

4.7 KNN

K Nearest Neighbors classifies a sample by computing the distance between the query point and all training samples. A common distance metric is the Euclidean distance:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^d (x_{ik} - x_{jk})^2} \quad (18)$$

After computing the distances, the algorithm selects the K closest neighbors and assigns the class label using majority voting

$$\hat{y} = \arg \max_c \sum_{i \in N_k(x)} I(y_i = c) \quad (19)$$

where $N_k(x)$ represents the set of the K nearest neighbors and $I(\cdot)$ is the indicator function.

4.8 Random Forest

Random forests are an extension of bagging methods that build an ensemble of decision trees using both bootstrap sampling and random feature selection. Each tree is trained on a different bootstrap sample of the dataset, and at each node, a random subset of features is considered to determine the best split. The training process consists of repeatedly generating bootstrap samples and growing decision trees by selecting the split that maximizes the information gain. The final prediction is obtained by aggregating the outputs of all trees using majority voting:

$$\hat{y} = \text{mode}\{C_1(x), C_2(x), \dots, C_m(x)\} \quad (20)$$

where $C_i(x)$ represents the prediction of the i -th decision tree.

4.9 Bagging

Bagging (Bootstrap Aggregating) is an ensemble method that reduces model variance by training multiple classifiers on different bootstrap samples of the training data. Each classifier is trained independently using a dataset generated by sampling with replacement from the original dataset. Similar to other ensemble methods, the final prediction is obtained through majority voting. However, unlike Random Forest, bagging does not introduce randomness in feature selection, and all features are considered when growing each decision tree.

4.10 Gradient Boosting

Gradient Boosting is another ensemble method that builds a strong prediction by calculating the error in the model before strategically adding weaker 'learner' models trained to predict the error in the main model. This is a sequential process, unlike Random Forest which is parallel, as it builds the full model iteratively and uses a learning rate to prevent overfitting. It uses gradient descent to minimize the error of the model as each new learner is added, which allows the model to reduce bias effectively.

5 Experimental Setup

5.1 Data Splitting

The dataset was split into training and testing sets using an 80/20 ratio. Stratified sampling was applied to preserve the class distribution across both sets. All model selection and hyperparameter tuning were performed exclusively on the training set, while the test set was reserved for final evaluation to ensure an unbiased estimate of model performance.

5.2 Model Training

A set of machine learning models was evaluated, including Logistic Regression, Support Vector Machines (linear and RBF kernels), k-Nearest Neighbors, Decision Tree, Random Forest, Bagging, Stochastic Gradient Descent, and Adaline. Each model was trained under consistent conditions using the same data splits and preprocessing configurations, ensuring a fair comparison across different models and dataset variants.

5.3 Hyperparameter Tuning

Hyperparameter tuning was performed using grid search combined with 5-fold cross-validation on the training set. The optimal model configuration was selected based on classification accuracy. The selected hyperparameters were then used to retrain each model on the full training set before evaluating performance on the test set.

5.4 Evaluation Metrics

Model performance was evaluated using accuracy, precision (macro-averaged), recall (macro-averaged), and F1-score (macro-averaged). Additionally, training accuracy, test accuracy, and cross-validation scores were recorded to assess model generalization and detect potential overfitting.

5.5 Implementation Details

The experimental pipeline was implemented in Python using the scikit-learn library. Experiments were conducted under reproducible conditions, using fixed random seeds and consistent data splits across all models.

6 Results and Discussions

The results in Table 1 show that model performance depends on both the model complexity and the preprocessing strategy. Ensemble methods, particularly Random Forest and Gradient Boosting, achieve the highest test performance, suggesting that nonlinear relationships exist in the data and are better captured by these models. However, the gap between training and testing accuracy indicates slight overfitting, which is expected for high-capacity models.

Dataset	Model	Train Accu	Test Accu
manual dropna	Random Forest	0.945	0.804
auto dropna	Logistic Regression	0.806	0.799
auto imputed	Logistic Regression	0.799	0.797
auto dropna	SVM Linear	0.806	0.796
manual dropna	Decision Tree	0.825	0.791

Table 1: Grid search results across best 5 models and dataset configurations

In contrast, Logistic Regression performs consistently well across all datasets, with small differences between training and testing metrics. This indicates good generalization and suggests that a significant portion of the data can be explained using linear decision boundaries. Bagging improves over a single decision tree by reducing variance, but it does not surpass Random Forest. This behavior reflects the importance of feature randomness in Random Forest, which allows it to better generalize. Similarly, SVM models show moderate performance, with the RBF kernel slightly outperforming the linear version, indicating the presence of some nonlinear patterns, although not dominant.

Several models exhibit overfitting, particularly those with near-perfect training accuracy such as decision trees and bagging. This is reflected in their lower test performance. In contrast, models with smaller train-test gaps, such as Logistic Regression, demonstrate

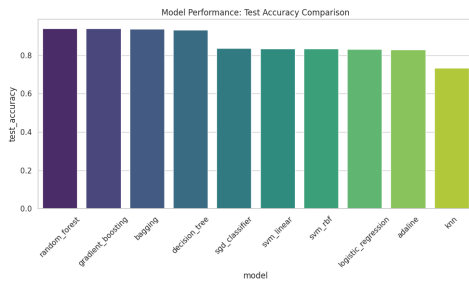


Figure 7: Test accuracy comparison Oasis data

better generalization, reinforcing the importance of evaluating models beyond training accuracy.

Preprocessing has also an impact on performance. Datasets constructed using automatic feature selection tend to produce more stable results, suggesting that reducing missing values is more beneficial than manually selecting features. Additionally, the comparison between imputed and non-imputed datasets reveals a trade-off: imputation increases dataset size, but may introduce noise, while removing incomplete samples can improve generalization.

Additionally, the Oasis dataset demonstrated that Random Forest and Gradient Boosting provide the most significant and efficient model for dementia classification. Furthermore Logistic Regression demonstrated a greater computational efficiency for rapid screening. In conclusion these findings show that transcriptomic and clinical biomarkers are effective at diagnosing clinical data when implemented correctly .

7 Future Work

Future work should focus on using more robust and larger datasets with well engineered features to evaluate whether improved data quality can further enhance model performance. Additionally, further hyperparameter tuning should be applied to the best-performing models, particularly ensemble methods and logistic regression, to obtain the optimal configuration for this dataset and improve generalization.

8 Conclusion

The results show that model performance is strongly influenced by both the choice of algorithm and the preprocessing strategy. Ensemble methods, particularly Random Forest and Gradient Boosting, achieve the best performance, indicating that nonlinear patterns are present in the data. However, these models also show signs of overfitting.

Logistic Regression performs consistently well across all datasets, with stable results and good generalization, showing that simpler models can remain competitive. Additionally, preprocessing plays a key role, as reducing missing values and feature selection to improve data quality leads to more reliable performance.

All models over all datasets struggled the most in classifying between MCI and CN. This is to be expected, as that is where the main problem lies and where uncertainty is high even in professional medical understanding today. However, the models proved quite successful in diagnosing active Alzheimer’s disease across

the datasets. This shows the potential diagnostic models have to identify early symptoms of Alzheimer’s and even further medical applications. With a well-engineered dataset and highly-optimized hyper-parameters for these models, many overlooked cases could be caught before diseases such as Alzheimer’s reach dangerous levels in patients.

9 Citations and Bibliographies

References

- [1] Andre Esteva, Alexandre Robicquet, Bharath Ramsundar, Volodymyr Kuleshov, Mark De Pisto, Katherine Chou, Claire Cui, Greg Corrado, Sebastian Thrun, and JeffDean. 2019. A guide to deep learning in healthcare. *Nature Medicine* 25 (Jan. 2019), 24–29. doi:10.1038/s41591-018-0316-z
- [2] Peter A. Lichtenberg. 2012. Misdiagnosis of Alzheimer’s Disease: Case Studies in Capacity Assessment. *Clinical Gerontologist* 35 (2012), 45–56. doi:10.1080/07317115.2011.626516
- [3] Philip Scheltens, Bart De Strooper, Miia Kivipelto, Henne Holstege, Gael Chéte-lat, Charlotte E Teunissen, Jeffrey Cummings, and Wiesje M van der Flier. 2021. Alzheimer’s disease. *Lancet* 397, 10284 (April 2021), 1577–1590. doi:10.1016/S0140-6736(20)32205-4

10 Acknowledgments

Generative AI was used for grammar, spelling, and minor clarity improvements. All technical content, analysis, and conclusions are the authors’ original work.

Received 31 March 2026; revised 31 March 2026; accepted 31 June 2026